

See also problems (06), (OR10).

(H1) (a) Are hyperbolic groups residually finite?

(b) Does every hyperbolic group have a proper subgroup of finite index ? [Background](#)

*(H2) Are hyperbolic groups linear? [Background](#)

*(H3) Do hyperbolic groups with torsion have solvable isomorphism problem? [Background](#)

(H4) (A.Miasnikov) Given a finite presentation of a hyperbolic group (which is not necessarily a Dehn presentation), is it possible to find a Dehn presentation for this group in polynomial time?

(H5) (A.Miasnikov) Given a finite presentation of an automatic group, can one decide if this group is hyperbolic? [Background](#)

(H6) (S.Gersten) Are all automatic groups biautomatic? [Background](#)

(H7) (S.Gersten) Does every automatic group have a solvable conjugacy problem? [Background](#)

(H8) (S.Gersten) Is every biautomatic group which does not contain any $\mathbb{Z} \times \mathbb{Z}$ subgroups, hyperbolic? [Background](#)

(H9) (S.Gersten) Can the group $\langle x, y; yxy^{-1} = x^2 \rangle$ be a subgroup of an automatic group? [Background](#)

(H10) (S.Gersten) Is a retract of an automatic group automatic? [Background](#)

(H11) Does every hyperbolic group act properly discontinuously and co-compactly by isometries on a $CAT(k)$ space, where $k < 0$?

(H12) (G.Baumslag, A.Miasnikov, V.Remeslennikov) Is every hyperbolic group equationally noetherian? (A group is called equationally noetherian if every system of equations in this group is equivalent to a finite subsystem). [Background](#)

(H13) Are combable groups automatic?

*(H14) (A.Miasnikov) We call a subgroup H of a group G malnormal if for any element g in G , but not in H , one has $H^g \cap H = \{1\}$. Is this property algorithmically decidable for finitely generated subgroups of hyperbolic groups? [Background](#)

(H15) (A.Miasnikov) If a finitely generated subgroup H of a hyperbolic group is malnormal (see above), does it follow that H is quasiconvex?

(H16) Is every metabelian automatic group virtually abelian ?

(H17) (A.Olshanskii) Does every hyperbolic group H have a free normal subgroup F such that the factor group H/F is of finite exponent?

(H18) (S.Gersten) Let F be Thompson's group:
 $(x_0, x_1, \dots \mid x_i x_k x_i^{-1} = x_{k+1}, k > i, k = 1, 2, \dots)$.
 Is F automatic? [Background](#)